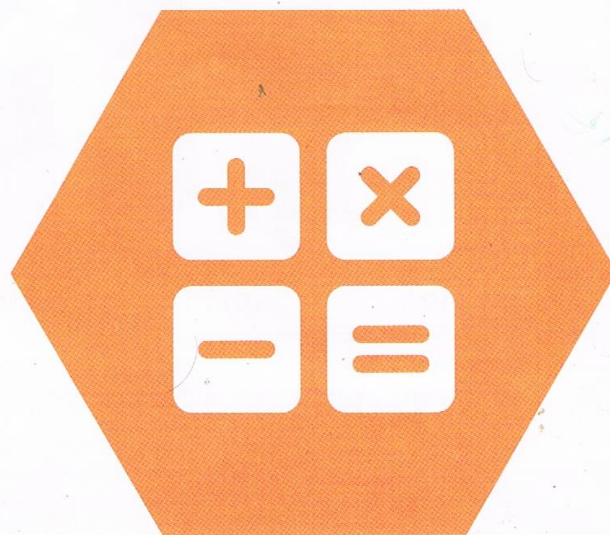




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

INTRODUCTORY PAPER



2014 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

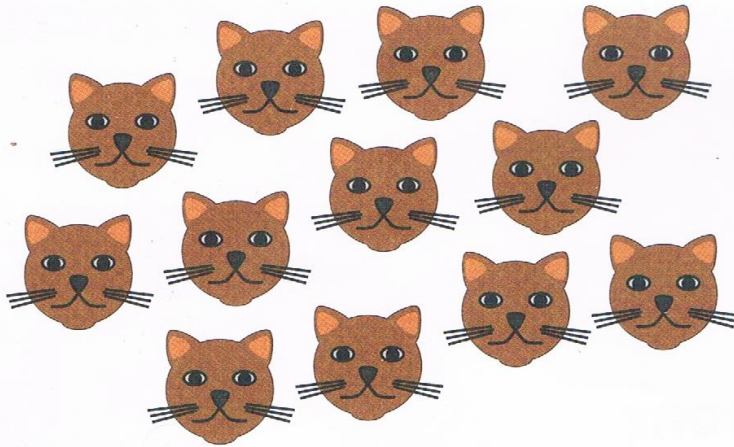
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

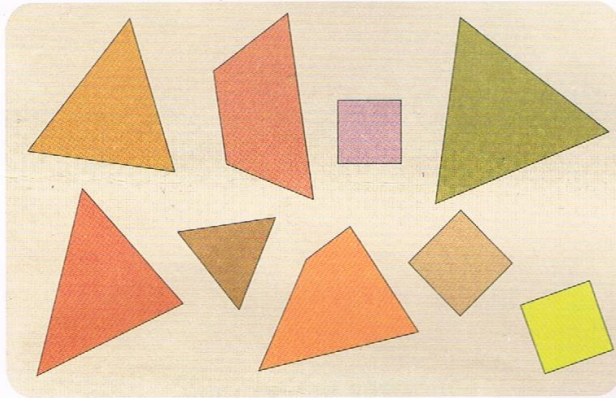
You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. How many cats are in this picture?



- (A) 10 (B) 11 (C) 12 (D) 13

2. How many triangles are in this picture?



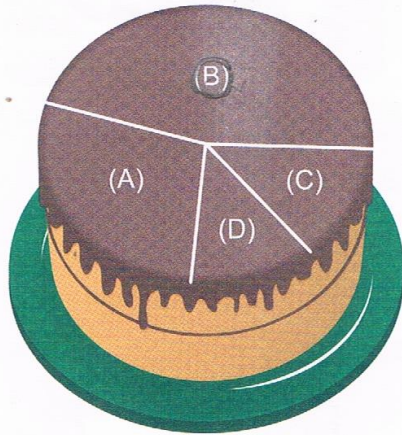
- (A) 4 (B) 5 (C) 7 (D) 8

3. If yesterday was Friday, what day is it tomorrow?

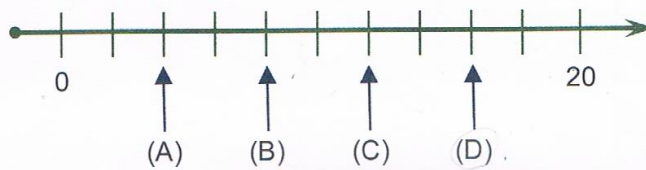
- (A) Thursday (B) Saturday (C) Sunday (D) Monday

4. Amy cut a cake into four unequal pieces.

Which piece was the largest?



5. Which arrow is pointing at 8 on this number line?



6. Sam drew a picture of a bee.



Which of these is also a picture of Sam's bee?



(A)



(B)



(C)



(D)

7. This pattern goes up by 6 each time.

8, 14, 20, 26, 32, ?

What is the next number in this pattern?

- (A) 33 (B) 34 (C) 36 (D) 38

8. This chart shows the shells some children collected at the beach.

Name	Shells
Peter	
Heidi	
Sue	
Henry	

How many shells did Sue collect altogether?

- (A) 4 (B) 5 (C) 6 (D) 8

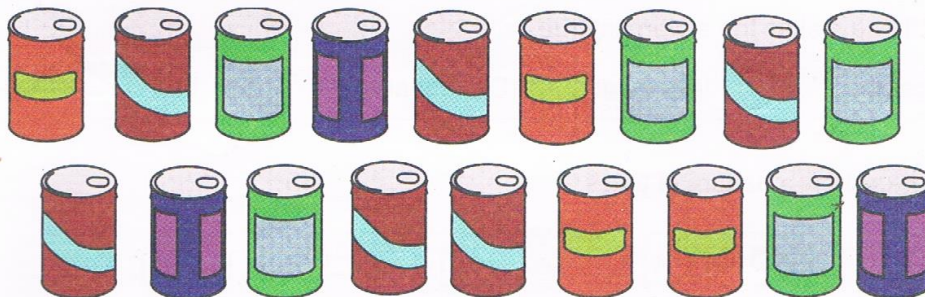
- 9.

 = 5  = 7

 +  +  +  +  = ?

- (A) 12 (B) 14 (C) 15 (D) 29

10. Ella picks one of these cans without looking.



Which kind of can is she most likely to pick?



(A)



(B)

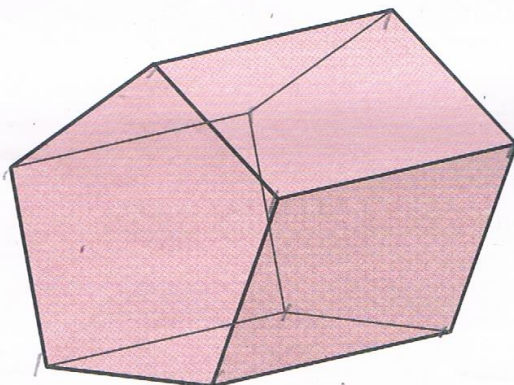


(C)



(D)

11. How many edges does this object have?



(A) 5

(B) 7

(C) 10

(D) 15

12. Jim drew some pictures.

Which of these is **impossible** for him to draw?

- (A) a cat with three legs
- (B) a square with five sides
- (C) a table with five legs
- (D) a car with three wheels

13. $24 - 9$ is $\overset{15}{17}$ $8 + 9$ $\overset{17}{17}$

Which of these makes the statement true?

- (A) bigger than (B) less than (C) equal to (D) half of

14. These clocks show a pattern of times.

10:30^{AM}

12:30^{PM}

?

04:30^{PM}

06:30^{PM}

Which of these clocks must replace the ? to fit the pattern?

11:30^{AM}

01:30^{PM}

02:30^{PM}

03:30^{PM}

(A)

(B)

(C)

(D)

15. Toby has 21 apples. Lillian has 4 apples less than Toby.

How many apples does Lillian have?

(A) 4

(B) 17

(C) 18

(D) 25

16. Sue is 180 cm tall. Mel is half as tall as Sue.

How tall is Mel?

(A) 40 cm

(B) 90 cm

(C) 100 cm

(D) 140 cm

17. $120 - 11 = ?$

(A) 131

(B) 119

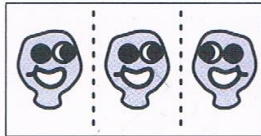
(C) 111

(D) 109

18. Sarah used this face to make a pattern. She flipped it over the dotted lines.



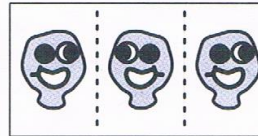
Which picture shows the result?



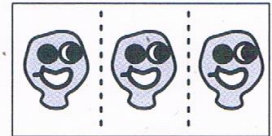
(A)



(B)



(C)



(D)

19. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15 (B) 16 (C) 31 (D) 56

20. Some students were walking along a track.

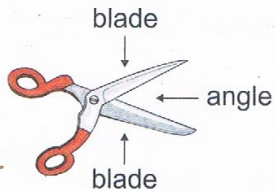
The table shows how far they had walked after 10 minutes.

Student	Distance
Sally	605 metres
Joe	575 metres
Nadia	520 metres
Billy	580 metres
Ellen	610 metres
Alex	565 metres

Who was halfway between Nadia and Ellen?

- (A) Sally (B) Joe (C) Billy (D) Alex

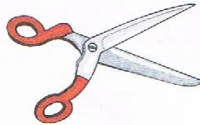
21. The blades of scissors can make an angle.



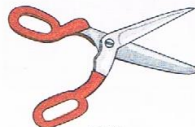
Which of these has the largest angle between its blades?



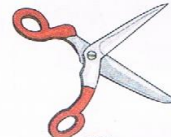
(A)



(B)

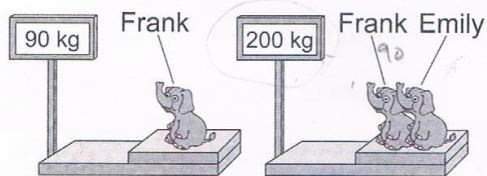


(C)



(D)

22. Emily and Frank are two baby elephants.

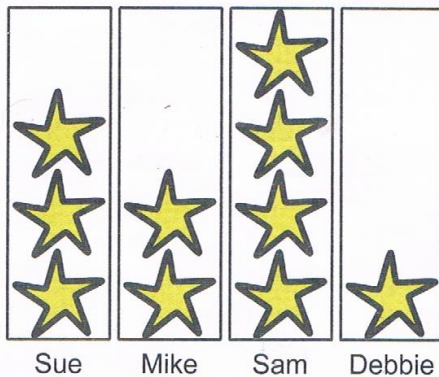


What is Emily's mass?

- (A) 90 kilograms (B) 110 kilograms (C) 190 kilograms (D) 200 kilograms

23. In Mr Zu's class you get a silver star for doing good work. If you have five silver stars, you get a gold star.

Gold Stars



Who has 15 silver stars?

- (A) Sue (B) Mike (C) Sam (D) Debbie

24.

$$\boxed{? + 15 = 28 - 8}$$

What is the missing number?

- (A) 5 (B) 10 (C) 20 (D) 35

25. Each week Anna saves \$10 of her pocket money.

Each week Tim gets \$16 pocket money and saves half of it.

In eight weeks, how much money do Anna and Tim save altogether?

- (A) \$208 (B) \$144 (C) \$88 (D) \$18

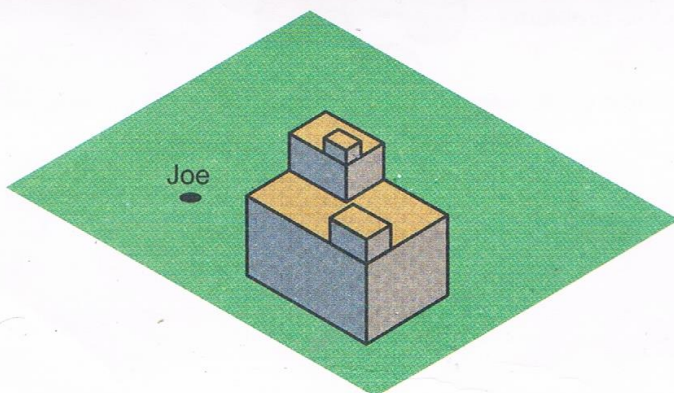
26. Abi is 1 year and 9 months older than Ben.

Ben is 2 years and 8 months younger than Cam.

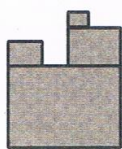
What is the difference in age between Abi and Cam?

- (A) 11 months (B) 13 months
(C) 4 years and 5 months (D) 4 years and 7 months

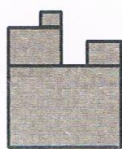
27. Joe is standing near a building as shown.



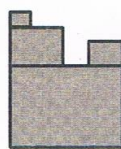
Which of these shows Joe's view of the building?



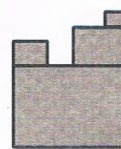
(A)



(B)



(C)

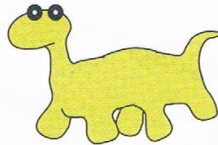


(D)

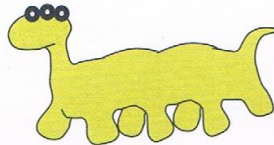
28. Jane is playing a video game. In the game there are monsters of different sizes.



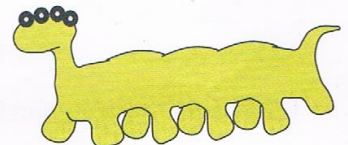
1 eye
3 legs



2 eyes
5 legs



3 eyes
7 legs



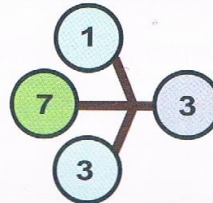
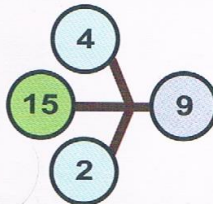
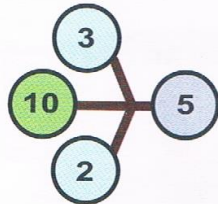
4 eyes
9 legs

Bigger monsters follow the same pattern.

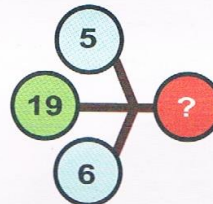
How many eyes would a monster with 25 legs have?

- (A) 12 (B) 13 (C) 14 (D) 15

29. Cindy has a special way of writing number problems.



What is the missing number in this problem?



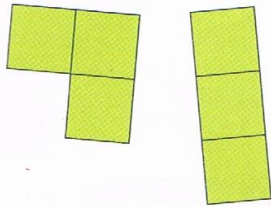
- (A) 8 (B) 11 (C) 13 (D) 14

30. Finn has some square tiles.

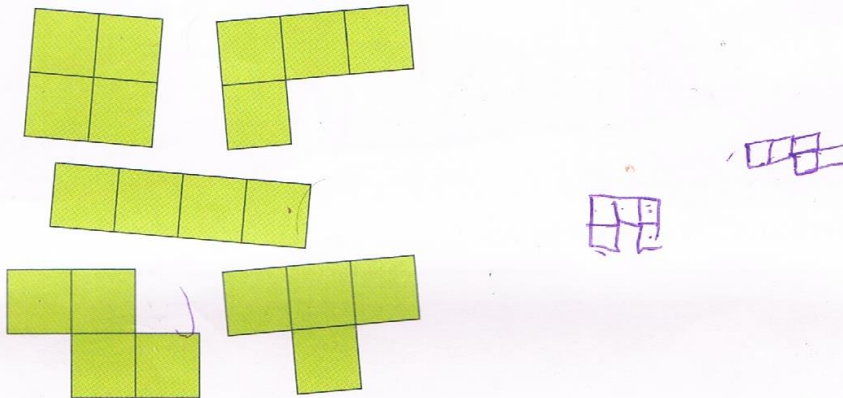
He uses two tiles together. He can only make this shape.



He uses three tiles together. He can only make these two different shapes.



He uses four tiles together. He can only make these five different shapes.



How many different shapes can Finn make using 5 tiles?

- (A) 6
- (B) 10
- (C) 12
- (D) 16

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PRACTICE QUESTION

What is the missing number?

$$2 + 3 = ?$$

- (A) 2
- (B) 3
- (C) 4
- ☒ (D) 5

Turn your answer sheet over and fill in the correct oval.

Australia comes
I got you and I

The following year levels should sit THIS Paper:

Australia	Year 2
Brunei	Primary 2
Indonesia	Year 3
Malaysia	Standard 2
New Zealand	Year 3
Pacific Region	Year 3
Singapore	Primary 1
South Africa	Grade 2



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ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

	Question content	Area assessed	Question number	Correct answer
Difficult Questions	Find the number of distinct arrangements that can be made with 5 square tiles	Space & Geometry	30	C
	Solve a problem reading a table and identifying a value halfway between two values	Number & Arithmetic	20	D
	Match a number to a position on a number line	Number & Arithmetic	5	B
	Identify the rule of an additive pattern	Algebra & Patterns	28	A
	Calculate in days, months and years	Measures & Units	26	A
	Identify days of the week	Measures & Units	3	C
	Find the missing number in a number statement	Algebra & Patterns	24	A
	Solve a word problem involving money	Number & Arithmetic	25	B
	Solve a number puzzle involving subtraction	Algebra & Patterns	29	A
	Compare an addition and a subtraction in terms of being greater or less in value	Number & Arithmetic	13	B
Easy Questions	Count the number of edges of a 3D object	Space & Geometry	11	D
	Find half of a three-digit number in context	Number & Arithmetic	16	B
	Identify objects after two flips (reflections)	Space & Geometry	18	C
	Identify which data displayed represents the a value	Chance & Data	23	A
	Solve a problem by comparing masses	Measures & Units	22	B
	Identify a view of an object from a given position	Space & Geometry	27	B
	Read and order digital clocks	Measures & Units	14	C
	Solve a subtraction in context	Number & Arithmetic	15	B
	Identify an impossible event	Chance & Data	12	B
	Subtract a two-digit number from a three-digit number	Number & Arithmetic	17	D
	Identify the relative size of an angle	Measures & Units	21	D
	Identify the most likely outcome when an object is picked from a set of objects	Chance & Data	10	A
	Represent an addition symbolically	Algebra & Patterns	9	D
	Read a table to solve a problem	Chance & Data	19	C
	Identify objects that are the same	Space & Geometry	6	A
	Predict the next number in a simple counting-on sequence	Algebra & Patterns	7	D
	Count the number of triangles in a picture	Number & Arithmetic	2	A
	Read a basic table	Chance & Data	8	C
	Identify the largest part of a whole cut into unequal parts	Measures & Units	4	B
	Identify (by counting) the number of objects in a collection	Number & Arithmetic	1	C